Locally toroidal families: application of the shared period proofs

An additive proof-location successor for the complete L121 contract

Private migration draft

The source is Fernandes–Leemans–Piedade–Ivić Weiss, *Two families of locally toroidal regular 4-hypertopes arising from toroids*, the retained twelve-page author edition. Its complete previously accepted application is preserved. The entire cubic, square, exact finite presentation and cubic-ambient proof sections now occur in the paid *Rotation certificates and affine period families* supplement, which itself uses the complete earlier marked-kernel common paper. Their original paragraph bytes and all whole computational dependencies are preserved in the migration map. This application makes no new mathematical or source-recovery claim.

For source Section 3, the shared cubic theorem identifies the literal presentation with E/sD for every $s \geq 3$, of order $48s^3$; it proves the stated ordered residues, intersection property, typed coset Helly, full regular group and faithful degree-$4s$ CPR action. The relation with the full cubic toroid is index two, not equality of the two groups. For Section 4, the complete square-family proof identifies the full marked affine quotient and proves the order $1152s^4$ and every source parameter, parabolic, flag and regularity assertion. These are uniform period arguments, not extrapolations from finite samples.

For Section 5, the exact shared finite theorem gives orders 720 and 46080, ordered maximal parabolic orders $(36, 48, 36, 48)$ and $(144, 48, 36, 48)$, and the degree-three local torus covering. Both modified presentations are identity-on-generators isomorphic to their respective originals. The full eleven-point countermodel and its proof are retained in the shared section: the modified relations do not force $(bd)^2 = 1$ and do not give order 240. All literal-source qualifications, including the extra parenthesis, remain those of the accepted predecessor.

The source's unrestricted minimum-degree suggestion remains separately qualified: $3D \cap 5D = 15D$ embeds $E/15D$ in $E/3D \times E/5D$. The disjoint union of the faithful degree 12 and 20 actions has degree 32, less than 60. It is intransitive; no minimum transitive degree or current open-question claim follows. No ordered local constraint or source correction is lost by referring to the full shared proofs.

The predecessor's complete source map, retained classical premises, program roles and six partial Lean declarations remain in force, with proof routes changed only to the shared supplement. No ratios have been calculated for this successor.